

Lecture 10: Long-Run Risks

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- Mehra-Prescott leaves too little equity premia unless risk aversion is pushed hard
- Habits and disasters fix this by changing risk prices or tail risk
- Bansal and Yaron ask a different question: can *small* persistent growth news move prices a lot?
- Long-run growth risk plus time-varying macro uncertainty is their answer
- With Epstein-Zin preferences, distant bad news becomes first-order for valuation

- Motivation: why long-run risk is not “just another $AR(1)$ ”
- Primitives: preferences, cash-flow dynamics, and state variables
- Approximate solution via the Campbell-Shiller decomposition ([Campbell and Shiller 1988](#))
- What must be solved numerically in the full model
- Quantitative targets, plus placeholders for the key figures and tables

Primitives

- Representative agent with recursive utility ([Epstein and Zin 1989, 1991](#))
- Two aggregate cash-flow objects: consumption and dividends
- The market is a claim to dividends, not a claim to aggregate consumption
- We therefore track both $R_{a,t+1}$ and $R_{m,t+1}$
- In the full model, the state vector is (x_t, σ_t^2)

$$\mathbb{E}_t \left[\delta^\theta G_{t+1}^{-\theta/\psi} R_{a,t+1}^{-(1-\theta)} R_{i,t+1} \right] = 1$$

- G_{t+1} is gross consumption growth
- ψ is the IES, γ is risk aversion
- $\theta \equiv \frac{1-\gamma}{1-1/\psi}$
- The whole mechanism needs the Epstein-Zin wedge between γ and ψ

Why Separate $R_{a,t+1}$ and $R_{m,t+1}$?

- $R_{a,t+1}$ is the return on a claim to aggregate consumption
- $R_{m,t+1}$ is the observable market return: the aggregate dividend claim
- Solve for the consumption claim first: it pins down the IMRS
- Then price the market using dividend dynamics
- This is where leverage in dividends will matter

$$x_{t+1} = \rho x_t + \varphi_e \sigma e_{t+1}$$

$$g_{t+1} = \mu + x_t + \sigma \eta_{t+1}$$

$$g_{d,t+1} = \mu_d + \phi x_t + \varphi_d \sigma u_{t+1}$$

- x_t is the small persistent component in expected growth
- $\phi > 1$ makes dividend cash flows more exposed than consumption

What Each Shock Does

- e_{t+1} moves beliefs about future growth
- η_{t+1} is short-run consumption risk
- u_{t+1} adds dividend risk not perfectly tied to consumption
- ρ close to one makes growth news long-lived
- Tiny news today can then have large valuation effects

$$\sigma_{t+1}^2 = \sigma^2 + \nu_1 (\sigma_t^2 - \sigma^2) + \sigma_w w_{t+1}$$

- Economic uncertainty is the conditional variance of consumption growth
- ν_1 controls how persistent volatility shocks are
- w_{t+1} becomes a priced shock in the full model
- This is what generates time-varying risk premia

Why This Is a Long-Run Risk Model

- The quantity shock is small
- The persistence is large
- Duration amplifies valuation responses to growth news
- If $\psi > 1$, good long-run growth news raises wealth and asset prices
- If $\psi < 1$, the same news works in the wrong direction

Approximate Solution

$$r_{a,t+1} = \kappa_0 + \kappa_1 z_{t+1} - z_t + g_{t+1}$$

- Lowercase letters are logs
- $z_t \equiv \log(P_t/C_t)$ is the log price-consumption ratio
- There is an analogous decomposition for the market using $z_{m,t}$
- This log-linearization makes the economics transparent

Guess the Price-Consumption Ratio

$$z_t = A_0 + A_1 x_t, \quad z_{m,t} = A_{0,m} + A_{1,m} x_t$$

- In Case I, one state variable is enough
- Plug the guess into the Euler equation and match coefficients
- In the full model, we add a loading on σ_t^2

$$A_1 = \frac{1 - 1/\psi}{1 - \kappa_1 \rho}, \quad A_{1,m} = \frac{\phi - 1/\psi}{1 - \kappa_{1,m} \rho}$$

- If $\psi > 1$, then $A_1 > 0$
- More persistent x_t raises both loadings
- If $\phi > 1$, dividend claims are more sensitive than consumption claims
- This is the core “small shock, big price effect” step

$$m_{t+1} = \theta \log \delta - \frac{\theta}{\psi} g_{t+1} + (\theta - 1) r_{a,t+1}$$

- Asset prices are about covariance with m_{t+1}
- Recursive utility makes the continuation return matter directly
- Long-run news enters the SDF through $r_{a,t+1}$

$$m_{t+1} - \mathbb{E}_t[m_{t+1}] = \lambda_{m,\eta}\sigma\eta_{t+1} - \lambda_{m,e}\sigma e_{t+1}$$

- $\lambda_{m,e}$ is the price of long-run growth news
- As ρ rises, growth news becomes more permanent
- More permanence means more covariance with the IMRS
- The long-run risk premium rises with persistence

$$z_t = A_0 + A_1 x_t + A_2 \sigma_t^2$$

- A_1 is unchanged
- A_2 captures how valuation reacts to economic uncertainty
- With $\gamma > 1$ and $\psi > 1$, the paper gets $A_2 < 0$
- Higher uncertainty lowers the wealth-consumption ratio

$$\mathbb{E}_t(r_{m,t+1} - r_{f,t}) = \beta_{m,e}\lambda_{m,e}\sigma_t^2 + \beta_{m,w}\lambda_{m,w}\sigma_w^2 - \frac{1}{2}\text{var}_t(r_{m,t+1})$$

- Growth-risk and volatility-risk both earn premia
- When σ_t is high, the conditional Sharpe ratio rises
- The model can produce high premia and a low, smooth risk-free rate at the same time

What the Approximation Teaches

- Long-run growth news matters because it moves future discount rates and future cash flows
- Volatility news matters because it changes risk premia and current valuations
- The sign restrictions hinge on $\psi > 1$
- Before any numerics, the Campbell-Shiller algebra already tells us what to look for

Quantitative Checks

Figure: Long-Horizon Return Predictability

Figure: Consumption Volatility and Valuation Ratios

Table: Baseline Calibration

Numerical Implementation

How to Solve It Numerically

- Treat (x_t, σ_t^2) as the state space
- Approximate policy functions or valuation ratios with a projection basis
- Choose coefficients to minimize Euler equation errors
- Simulate at the monthly decision frequency
- Time-aggregate simulated data to annual moments before comparing to the data

How to Solve It Numerically

- Treat (x_t, σ_t^2) as the state space
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- Time-aggregate simulated data to annual moments before comparing to the data
- In the paper, the numerical results are close to the log-linear analytics

What to Match

- Mean and persistence of consumption and dividend growth
- Equity premium and average risk-free rate
- Volatility of the market return, the risk-free rate, and the price-dividend ratio
- Long-horizon return predictability
- The negative relation between economic uncertainty and valuation ratios

Takeaways

- The paper prices *persistent* growth news, not just bad short-run shocks
- Epstein-Zin preferences are doing essential work: $\psi > 1$ is not optional here
- Stochastic volatility gives the model time-varying premia and volatility feedback
- The approximation is simple enough to teach, but the full quantitative model is numerical
- Next step in class: decide which empirical moments we want to emphasize

Questions?

Readings and References i

- Main paper: Bansal and Yaron (2004), *Risks for the Long Run: A Potential Resolution of Asset Pricing Puzzles*
- Campbell-Shiller log-linearization (Campbell and Shiller 1988)
- Recursive utility foundations (Epstein and Zin 1989, 1991)
- Useful benchmark for comparison: habits (Campbell and Cochrane 1999)
- Campbell, Chapter 6

Campbell, John Y., and John H. Cochrane. 1999. "By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior." *Journal of Political Economy* 107 (2): 205–51.
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